Help Desk Ticket Performance Analysis

End-to-End Data Analytics Project using SQL, Python & Power BI

Business Domain: IT Service Management (ITSM)

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Project Summary

This project presents an end-to-end analysis of a Help Desk Ticket dataset to understand operational performance and identify opportunities for improving service efficiency. The analysis covers data cleaning, exploratory data analysis, SQL-based business analysis, and the development of an interactive Power BI dashboard.

The project focuses on understanding ticket volume, resolution performance, reassignment behavior, ticket priority, impact levels, and category-wise performance. The final dashboard enables users to explore these metrics through interactive filters and visualizations, providing insights that can support operational decision-making.

Tools & Technologies

- SQL (MySQL)
- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Power BI
- DAX

Project Duration: Self-Learning Portfolio Project

Dataset: Service Now Help Desk Incident Dataset

Project Workflow:

Raw Dataset → Data Cleaning → EDA → SQL Analysis → Power BI Dashboard → Business Insights

1. Project Overview

Efficient help desk operations play an important role in maintaining the quality of IT services within an organization. Every support ticket represents a user issue that requires timely resolution, and analyzing these tickets can help identify operational bottlenecks, recurring problems, and areas where service performance can be improved.

In this project, I performed an end-to-end analysis of a ServiceNow help desk incident dataset containing over 140,000 incident records. The project combines SQL, Python, and Power BI to transform raw operational data into meaningful business insights through data cleaning, exploratory analysis, SQL-based reporting, and an interactive dashboard.

The analysis focuses on understanding ticket volume, resolution efficiency, reassignment patterns, priority distribution, impact levels, and category-wise performance. The final dashboard provides an interactive view of these metrics, allowing users to filter and explore the data from different operational perspectives.

Rather than focusing only on descriptive statistics, the project aims to answer practical business questions that could help an IT support team monitor performance, identify recurring issues, and make data-driven operational decisions.

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2. Business Problem

Help desk teams often manage thousands of support tickets every month, making it difficult to identify operational issues by simply reviewing raw records. Without proper analysis, organizations may struggle to understand which categories generate the highest workload, which incidents take the longest to resolve, or how factors such as priority and ticket reassignment influence overall service performance.

This project was carried out to analyze historical incident data and convert it into meaningful operational insights. The objective was not only to measure key performance indicators but also to identify patterns that could support better resource allocation and process improvement.

The analysis was designed to answer questions such as:

- How does ticket volume change over time?
- Which ticket categories generate the highest workload?
- Which categories require the longest average resolution time?
- How does ticket priority affect resolution time?
- Does increasing ticket reassignment lead to longer resolution times?
- What is the overall SLA failure rate?
- How are incidents distributed across different impact levels?

The answers to these questions provide a clearer understanding of help desk performance and demonstrate how data analysis can support operational decision-making.

3. Dataset Description

The analysis is based on a ServiceNow Help Desk Incident dataset containing historical incident records generated by an IT support system. Each record represents a single incident ticket and includes information related to ticket creation, priority, impact, assignment details, resolution time, SLA status, and ticket lifecycle.

The dataset covers more than 140,000 incident records collected over one year. Since it originated from an operational system, preprocessing was required before analysis.

The dataset includes attributes describing different stages of the incident management process, such as:

- Ticket creation and closure timestamps
- Category and subcategory of incidents
- Assignment group responsible for resolution
- Priority and impact levels
- Reassignment count
- SLA status
- Resolution time
- Ticket state

These attributes enable trend analysis, operational performance evaluation, and service efficiency measurement.

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4. Data Preparation and Cleaning

Before starting the analysis, the dataset was cleaned to improve consistency and ensure that the calculated metrics were reliable.

The preprocessing stage was performed using Python with the Pandas library. During this stage, unnecessary columns were removed, missing values were examined, and appropriate data types were assigned to each variable.

Timestamp columns were converted into datetime format to enable time-based analysis. From these timestamps, additional features such as month and year were created to support trend analysis in both SQL and Power BI.

A new **Resolution Hours** column was calculated by measuring the time difference between ticket creation and ticket closure. This metric became one of the primary performance indicators used throughout the project.

Some categorical fields contained missing values, particularly the assignment group information. Instead of removing these incident records, the missing values were handled separately to preserve the integrity of the dataset while ensuring cleaner reporting in the dashboard.

After completing the preprocessing stage, the cleaned dataset was exported and used consistently for SQL analysis, exploratory data analysis, and Power BI dashboard development.

5. Exploratory Data Analysis (Python)

After cleaning the dataset, Exploratory Data Analysis (EDA) was performed using Python to understand the overall structure of the data and identify meaningful patterns before carrying out SQL analysis and dashboard development.

The analysis included summary statistics, distribution analysis, missing value assessment, and visual exploration of both numerical and categorical variables. Libraries such as Pandas, Matplotlib, and Seaborn were used to generate statistical summaries and visualizations.

The EDA phase focused on answering questions such as:

- How are tickets distributed across different categories?
- Which priority levels occur most frequently?
- How are impact levels distributed?
- How does resolution time vary across different priorities?
- Are there any unusual values or outliers in the numerical variables?
- Which variables appear to influence operational performance?

Several visualizations, including bar charts, count plots, box plots, histograms, correlation heatmaps, and trend charts, were used to examine the relationships between variables and identify patterns that required further investigation.

This stage also helped validate the quality of the cleaned dataset before importing it into SQL and Power BI. Any observations identified during EDA were later verified through SQL queries and incorporated into the final dashboard to ensure consistency across the entire analysis workflow.

Rather than drawing final business conclusions during this stage, the primary objective of EDA was to understand the characteristics of the dataset, identify potential trends, and guide the subsequent analytical process.

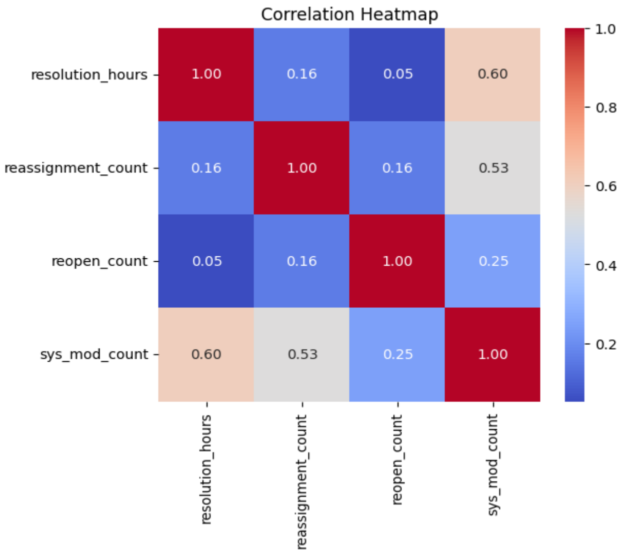

Figure 1. Correlation Heatmap

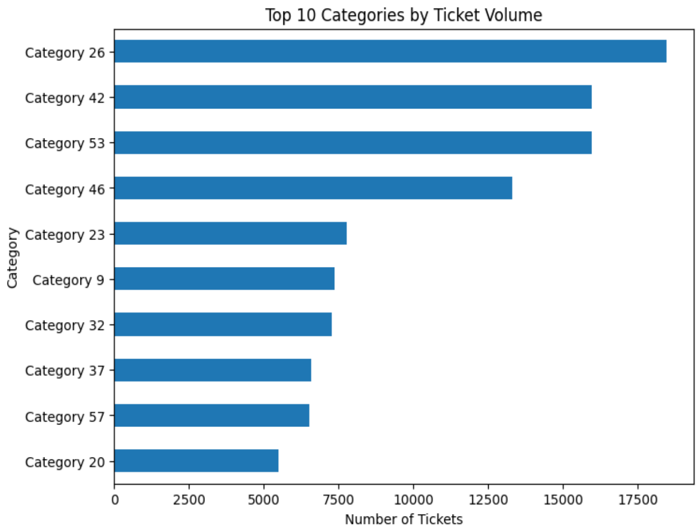

Figure 2. Top 10 Categories by Ticket Volume

6. SQL Analysis

After completing data cleaning and exploratory analysis, SQL was used to perform business-focused analysis on the cleaned dataset. The objective was to convert raw incident records into meaningful performance metrics that could support operational decision-making.

The SQL analysis was organized into multiple sections, with each group of queries answering a specific business question instead of simply retrieving data.

Ticket Volume Analysis

The first stage focused on understanding the overall workload handled by the help desk. SQL queries were used to calculate the total number of incident tickets, analyze monthly ticket trends, and identify the categories generating the highest ticket volumes. These results helped highlight periods of increased workload and the areas contributing most to support demand.

Resolution Performance Analysis

The second stage evaluated operational efficiency by calculating the average resolution time for incident tickets. Additional queries compared resolution time across different priority levels and analyzed how ticket reassignment influenced the overall resolution process. This helped identify factors associated with longer ticket resolution times.

SLA Performance Analysis

Service Level Agreement (SLA) compliance was measured by calculating the overall SLA failure rate, which was found to be 6.5%. This indicates that 93.5% of tickets were resolved within the defined service level target, reflecting strong baseline compliance despite high overall ticket volume.

Category-Level Analysis

Category-level analysis was performed to identify which incident categories generated the highest workload and which categories required the longest average resolution time. Ranking these categories made it easier to identify recurring operational issues that could benefit from process improvements or additional support resources.

Operational Summary

The SQL analysis transformed raw transactional data into structured business metrics that were later used during dashboard development. The calculated measures provided the foundation for the KPI cards, trend analysis, and category-level visualizations presented in the Power BI dashboard.

Rather than using SQL only for data retrieval, it served as the primary analytical tool for validating business metrics, answering operational questions, and supporting the insights presented throughout this project.

7. Power BI Dashboard

The final stage of the project involved developing an interactive Power BI dashboard to present the analytical findings in a clear and user-friendly format. The dashboard was designed to provide a consolidated view of help desk performance, allowing users to monitor key operational metrics and explore the data through interactive filters.

The dashboard combines KPIs, trend analysis, and category-level insights into a single interactive interface for monitoring operational performance.

Dashboard Components

Key Performance Indicators (KPIs)

The top section of the dashboard presents four primary performance indicators that summarize the overall state of the help desk operation.

- **Total Tickets** – Displays the total number of incident tickets available in the dataset.
- **Average Resolution Time** – Measures the average time required to resolve an incident.
- **SLA Failure Rate** – Indicates the percentage of incidents that exceeded the defined service level agreement.
- **Average Reassignment Count** – Shows how frequently tickets were reassigned before resolution.

These KPIs provide a quick overview of workload, service efficiency, and operational performance.

Trend Analysis

The monthly ticket volume chart illustrates how incident demand changed throughout the reporting period. Monitoring ticket trends helps identify periods of increased workload and supports better resource planning.

Impact Distribution

The impact-level visualization shows how incidents are distributed across different business impact categories. This helps determine whether the majority of support requests involve low, medium, or high business impact.

Resolution Time Analysis

Average resolution time is analyzed from multiple perspectives to understand operational efficiency.

- Resolution time by **Priority** highlights how service performance differs across ticket priorities.
- Resolution time by **Category** identifies the categories requiring the longest average time to resolve.
- Resolution time by **Reassignment Count** illustrates the relationship between ticket reassignments and resolution efficiency.

Together, these visualizations help identify process bottlenecks and factors associated with longer ticket resolution times

Ticket Category Analysis

The dashboard also highlights the categories generating the highest ticket volumes. Ranking categories by ticket count helps identify recurring operational issues and supports prioritization of improvement efforts.

Interactive Filtering

Slicers for **Month**, **Priority**, and **Assignment Group** allow users to filter the dashboard dynamically. This enables analysis from different operational perspectives without modifying the underlying data or calculations.

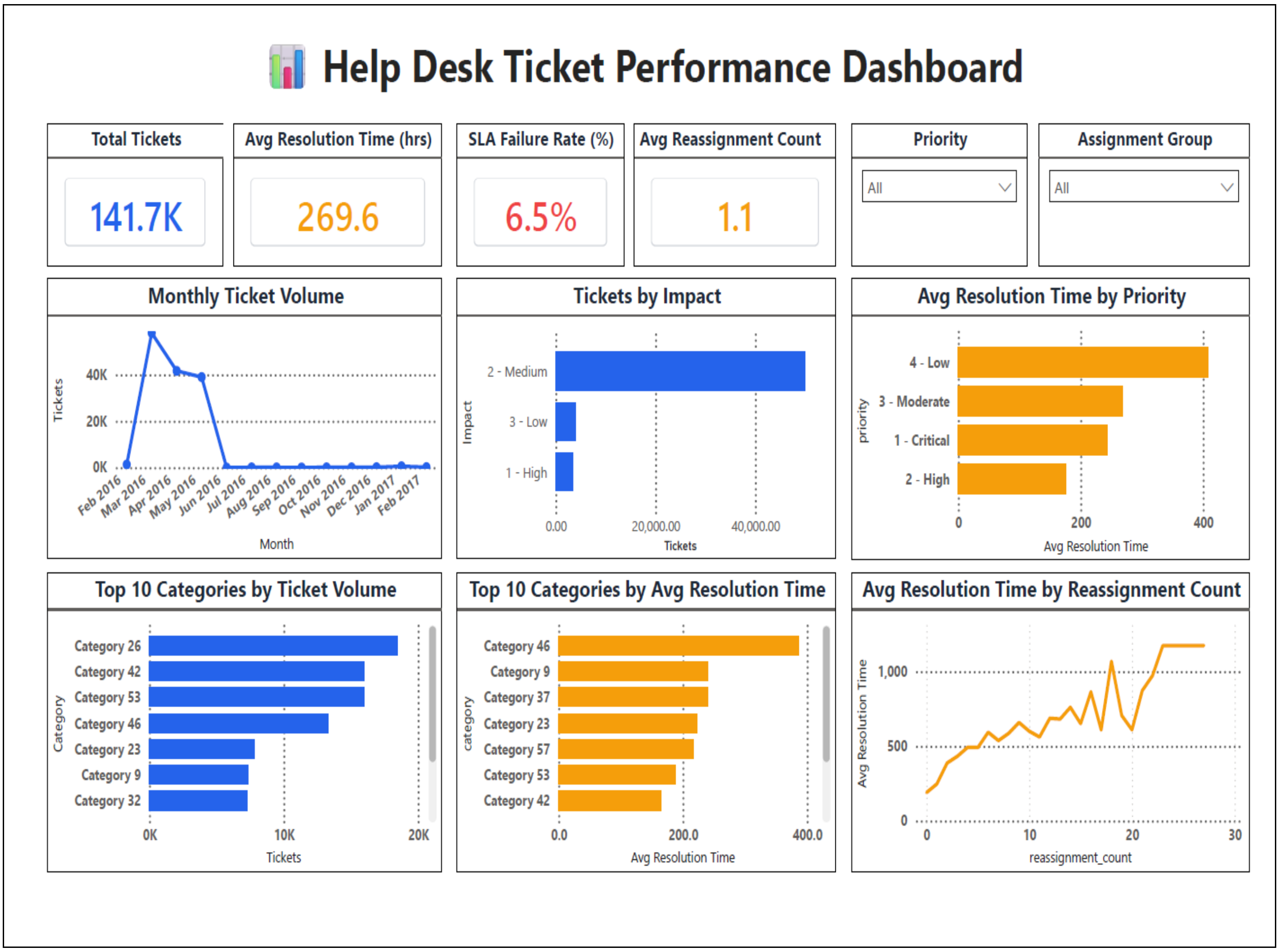

Figure 3. Final Power BI Dashboard

8. Key Business Insights

8.1 Help Desk Handles High Volume with Strong SLA Compliance

The dataset captures 141.7K incident tickets with an overall SLA failure rate of just 6.5%, meaning 93.5% of tickets were resolved within target. However, the overall average resolution time stands at 269.6 hours (~11.2 days), suggesting that while most tickets clear SLA thresholds, a long tail of slow-moving tickets significantly inflates the average.

8.2 Ticket Volume Spiked Sharply In Early 2016

Monthly ticket volume peaked in March 2016 at 42K tickets, following a steep rise from February, then settled into a stable range of under 1K tickets per month from mid-2016 through early 2017. This pattern likely reflects a data migration or system go-live event rather than organic demand, and is an important caveat when interpreting trend data.

8.3 Low-Priority Tickets Take Significantly Longer

Avg resolution time by priority reveals a counterintuitive pattern: 4-Low priority tickets averaged 408 hours, more than double the 176 hours for 2-High priority tickets, and well above the 243 hours for 1-Critical tickets. This indicates the support team correctly fast-tracks urgent issues, but low-priority tickets are being deprioritized and left unresolved for extended periods, creating a hidden backlog risk.

8.4 Reassignment Count Is the Strongest Driver of Delay

Resolution time rises sharply with reassignment count, from 190 hours at 0 reassignments to over 1,174 hours once tickets are reassigned 25+ times. Even a single reassignment more than triples resolution time (190 to 594 hours), making reassignment frequency one of the clearest operational levers for improving turnaround.

8.5 A Concentrated Set of Categories Drives Most of the Workload

Category 26 alone generated 18K tickets, the highest of any category, followed closely by Category 42 (16K) and Category 53 (16K). Together, the top 3 categories account for roughly 35% of total ticket volume, confirming that a small number of recurring issue types are responsible for the bulk of help desk demand.

8.6 The Slowest Categories to Resolve Are Not the Highest-Volume Ones

Category 46 had the longest average resolution time at 386.4 hours, despite ranking only 4th in ticket volume (13K tickets). This category appears in both the top-volume and top-resolution-time lists, making it the strongest candidate for root-cause investigation — it is both high-traffic and operationally slow.

8.7 Medium-Impact Incidents Dominate the Workload

134,330 tickets (roughly 95% of all tickets) were classified as 2-Medium impact, compared to just 3,886 Low-impact and 3,491 High-impact tickets. This shows the help desk's workload is overwhelmingly driven by routine operational issues rather than critical business-impacting incidents, which should inform staffing and triage policy.

8.8 Interactive Reporting Improved Operational Visibility

By combining SQL-derived KPIs with an interactive Power BI dashboard, stakeholders can filter performance by priority and assignment group across 141.7K records, enabling faster identification of bottlenecks like reassignment-driven delays and slow-resolving categories without manual review.

9. Business Recommendations

Based on the findings from the analysis, the following recommendations could help improve the overall efficiency of the help desk operation.

9.1 Reduce Unnecessary Ticket Reassignments

Reassignment count emerged as the strongest driver of delay in the dataset — average resolution time rose from 190 hours at 0 reassignments to over 1,174 hours once tickets were reassigned 25+ times, with even a single reassignment more than tripling resolution time. Establishing clearer ownership rules and improving first-touch ticket routing could significantly reduce this escalation pattern.

9.2 Investigate Category 46 and Other High-Resolution-Time Categories

Category 46 recorded the longest average resolution time at 386.4 hours, despite ranking only 4th in ticket volume, making it the clearest candidate for root-cause investigation. A focused review of this category's workflow, required expertise, and common blockers could meaningfully reduce overall average resolution time.

9.3 Monitor High-Volume Categories

Category 26 generated nearly 18K tickets, the highest volume in the dataset, with Category 42 and Category 53 close behind at roughly 16K each — together accounting for around 35% of total ticket volume. Targeting process improvements at these three categories has the potential to meaningfully reduce overall workload.

9.4 Continue Monitoring SLA Performance

The overall SLA failure rate was 6.5%, indicating that most tickets met the defined service level target. This metric should continue to be tracked closely, since the high average resolution time (269.6 hours) suggests a long tail of slow tickets that could push the failure rate higher if left unaddressed.

9.5 Review Prioritization of Low-Priority Tickets

Low-priority tickets averaged 408 hours to resolve, more than double the 176 hours for High-priority tickets. While this reflects correct triage of urgent issues, it also signals a growing backlog risk in low-priority queues that could affect long-term service quality if these tickets are consistently deprioritized.

9.6 Use Dashboard Insights for Operational Planning

The interactive dashboard enables managers to monitor ticket trends, compare operational metrics, and evaluate performance across different priorities, assignment groups, and time periods. Using these insights during regular operational reviews can support more informed, data-backed decision-making.

10. Conclusion

This project demonstrates a complete end-to-end data analytics workflow, beginning with raw operational data and ending with an interactive business dashboard. Python was used to clean and prepare the dataset, exploratory data analysis was performed to understand the data, SQL was used to answer business questions, and Power BI was used to present the results through interactive visualizations.

The analysis identified important patterns related to ticket volume, resolution efficiency, reassignment behaviour, priority levels, impact distribution, and category-level performance. These findings show how historical incident data can be transformed into meaningful business insights that support operational monitoring and process improvement.

Beyond developing technical skills in SQL, Python, and Power BI, this project strengthened my ability to approach a business problem systematically, interpret analytical results, and communicate findings through clear visualizations and structured reporting.

11. Future Improvements

This project can be extended in several ways to provide additional business value.

- Develop a machine learning model to predict ticket resolution time.
- Forecast future ticket volume using time series analysis.
- Build an automated data pipeline to refresh the dashboard periodically.
- Include additional operational metrics such as first response time and customer satisfaction scores if available.
- Publish the dashboard using Power BI Service for real-time monitoring and easier collaboration.